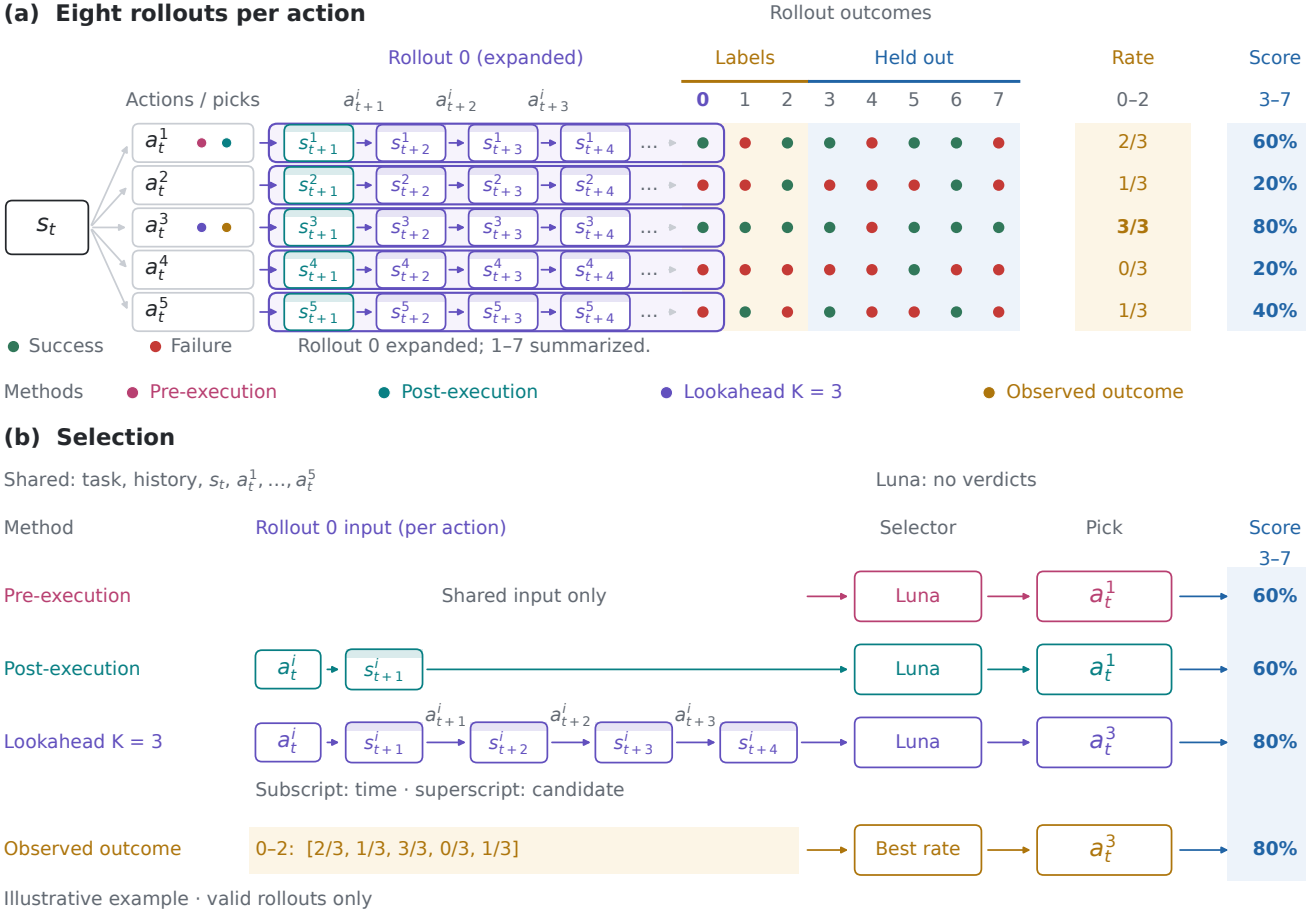
(a) Eight rollouts per action
Rollout outcomes
Rollout 0 (expanded)
Labels
Held out
Actions / picks
0 1 2 3 4 5 6 7
Rate
0–2
Score
3–7
s_t
a_t^1 2/3 60%
a_t^2 1/3 20%
a_t^3 3/3 80%
a_t^4 0/3 20%
a_t^5 1/3 40%
Success
Failure
Rollout 0 expanded; 1–7 summarized.
Methods
Pre-execution
Post-execution
Lookahead K = 3
Observed outcome
(b) Selection
Shared: task, history, s_t, a_t^1, …, a_t^5
Luna: no verdicts
Method
Rollout 0 input (per action)
Selector
Pick
Score
3–7
Pre-execution
Shared input only
Luna
a_t^1
60%
Post-execution
a_t^i s_{t+1}^i
Luna
a_t^1
60%
Lookahead K = 3
a_t^i s_{t+1}^i a_{t+1}^i s_{t+2}^i a_{t+2}^i s_{t+3}^i a_{t+3}^i s_{t+4}^i
Luna
a_t^3
80%
Subscript: time · superscript: candidate
Observed outcome
0–2: [2/3, 1/3, 3/3, 0/3, 1/3]
Best rate
a_t^3
80%
Illustrative example · valid rollouts only